



University of Piraeus
School of Information and Communication Technologies
Department of Digital Systems

Level: Master's Programme of Studies

Network Security

Case 5

Supervising Professor: Prof. Christos Xenakis

Ioannis Kalaitzidis

ioannis_kalaitzidis@ssl-unipi.gr

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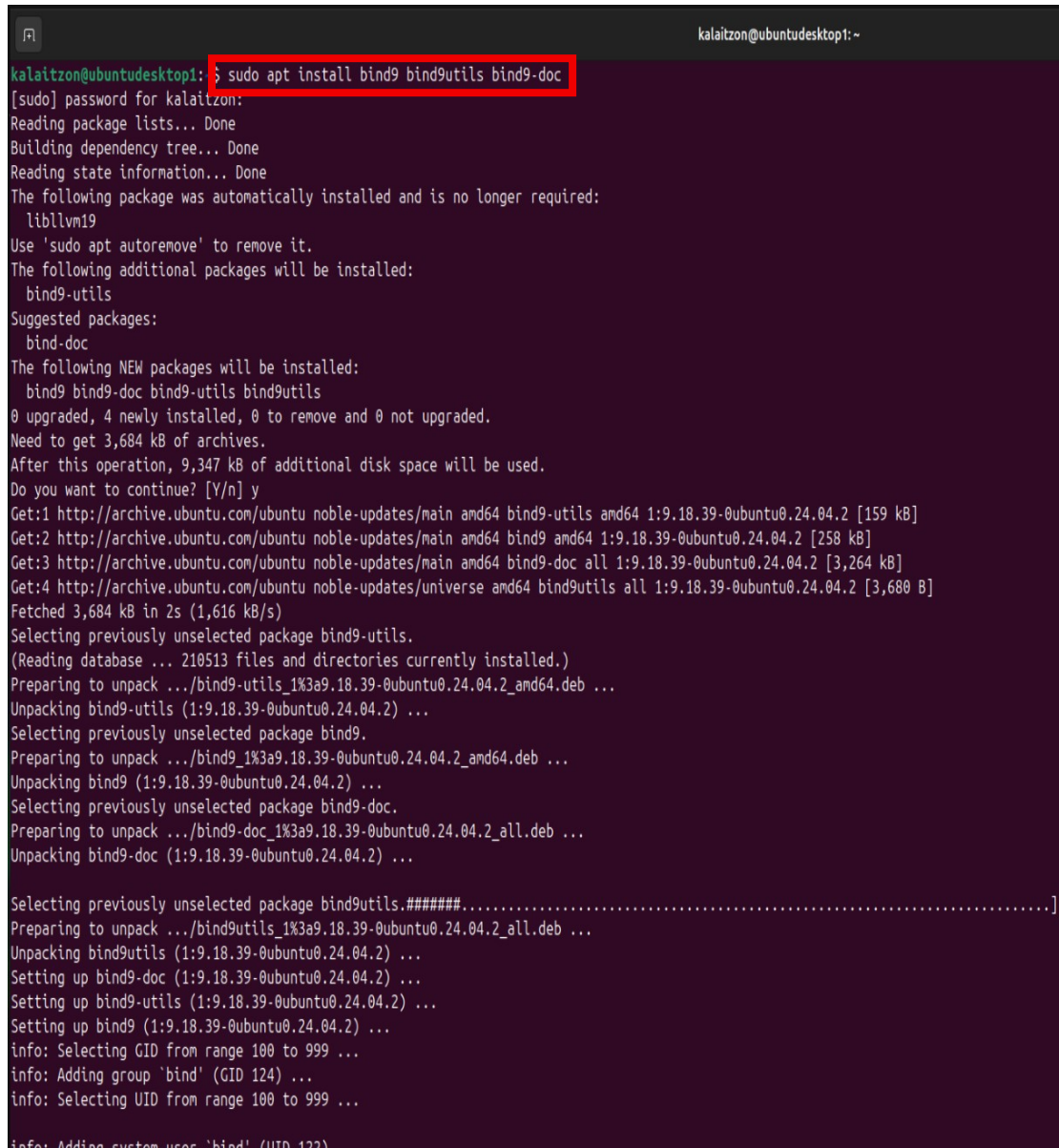
Abstract

Installation of a DNS server using the BIND9 tool on Ubuntu and creation of a Master Zone for the 'Kalaitzon.gr' domain. Configuration of the Forwarding function to Google and mapping of the domain to the server's IP. Correct operation was verified through testing from a Windows 10 (client).

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Practical part of the assignment

A terminal window with a dark purple background and white text. The prompt is 'kalaizon@ubuntu0.24.04.2: ~'. The command 'sudo apt install bind9 bind9utils bind9-doc' is entered and highlighted with a red box. The terminal shows the progress of the installation, including package lists, dependency trees, and disk space requirements. It lists additional packages to be installed (bind9-doc, bind9-utils) and suggested packages (bind-doc). It shows the download of four packages from the Ubuntu archive, totaling 3,684 kB. The installation process is shown step-by-step, including unpacking and setting up each package. The final output shows the group 'bind' being added and the user 'bind' being created.

```
kalaizon@ubuntu0.24.04.2: ~$ sudo apt install bind9 bind9utils bind9-doc
[sudo] password for kalaizon:
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following package was automatically installed and is no longer required:
  libllvm19
Use 'sudo apt autoremove' to remove it.
The following additional packages will be installed:
  bind9-utils
Suggested packages:
  bind-doc
The following NEW packages will be installed:
  bind9 bind9-doc bind9-utils bind9utils
0 upgraded, 4 newly installed, 0 to remove and 0 not upgraded.
Need to get 3,684 kB of archives.
After this operation, 9,347 kB of additional disk space will be used.
Do you want to continue? [Y/n] y
Get:1 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 bind9-utils amd64 1:9.18.39-0ubuntu0.24.04.2 [159 kB]
Get:2 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 bind9 amd64 1:9.18.39-0ubuntu0.24.04.2 [258 kB]
Get:3 http://archive.ubuntu.com/ubuntu noble-updates/main amd64 bind9-doc all 1:9.18.39-0ubuntu0.24.04.2 [3,264 kB]
Get:4 http://archive.ubuntu.com/ubuntu noble-updates/universe amd64 bind9utils amd64 1:9.18.39-0ubuntu0.24.04.2 [3,680 B]
Fetched 3,684 kB in 2s (1,616 kB/s)
Selecting previously unselected package bind9-utils.
(Reading database ... 210513 files and directories currently installed.)
Preparing to unpack .../bind9-utils_1%3a9.18.39-0ubuntu0.24.04.2_amd64.deb ...
Unpacking bind9-utils (1:9.18.39-0ubuntu0.24.04.2) ...
Selecting previously unselected package bind9.
Preparing to unpack .../bind9_1%3a9.18.39-0ubuntu0.24.04.2_amd64.deb ...
Unpacking bind9 (1:9.18.39-0ubuntu0.24.04.2) ...
Selecting previously unselected package bind9-doc.
Preparing to unpack .../bind9-doc_1%3a9.18.39-0ubuntu0.24.04.2_all.deb ...
Unpacking bind9-doc (1:9.18.39-0ubuntu0.24.04.2) ...
Selecting previously unselected package bind9utils.#####
Preparing to unpack .../bind9utils_1%3a9.18.39-0ubuntu0.24.04.2_all.deb ...
Unpacking bind9utils (1:9.18.39-0ubuntu0.24.04.2) ...
Setting up bind9-doc (1:9.18.39-0ubuntu0.24.04.2) ...
Setting up bind9-utils (1:9.18.39-0ubuntu0.24.04.2) ...
Setting up bind9 (1:9.18.39-0ubuntu0.24.04.2) ...
info: Selecting GID from range 100 to 999 ...
info: Adding group 'bind' (GID 124) ...
info: Selecting UID from range 100 to 999 ...
info: Adding system user 'bind' (UID 122)
```

We 'download' bind9 with the command `sudo apt install bind9 bind9utils bind9-doc`.

Next, we configure Forwarding (Recursive Query). Without this, our server can only answer queries for our own domain (e.g. Kalaitzon.gr). With the command `sudo nano /etc/bind/named.conf.options` we open the settings file and remove the comment (`//`), entering Google's IP (8.8.8.8, which is normally 008.008.008.008). We save and exit by pressing `Ctrl + O` and `Enter`. Then `Ctrl + X` to return to the terminal. We do this

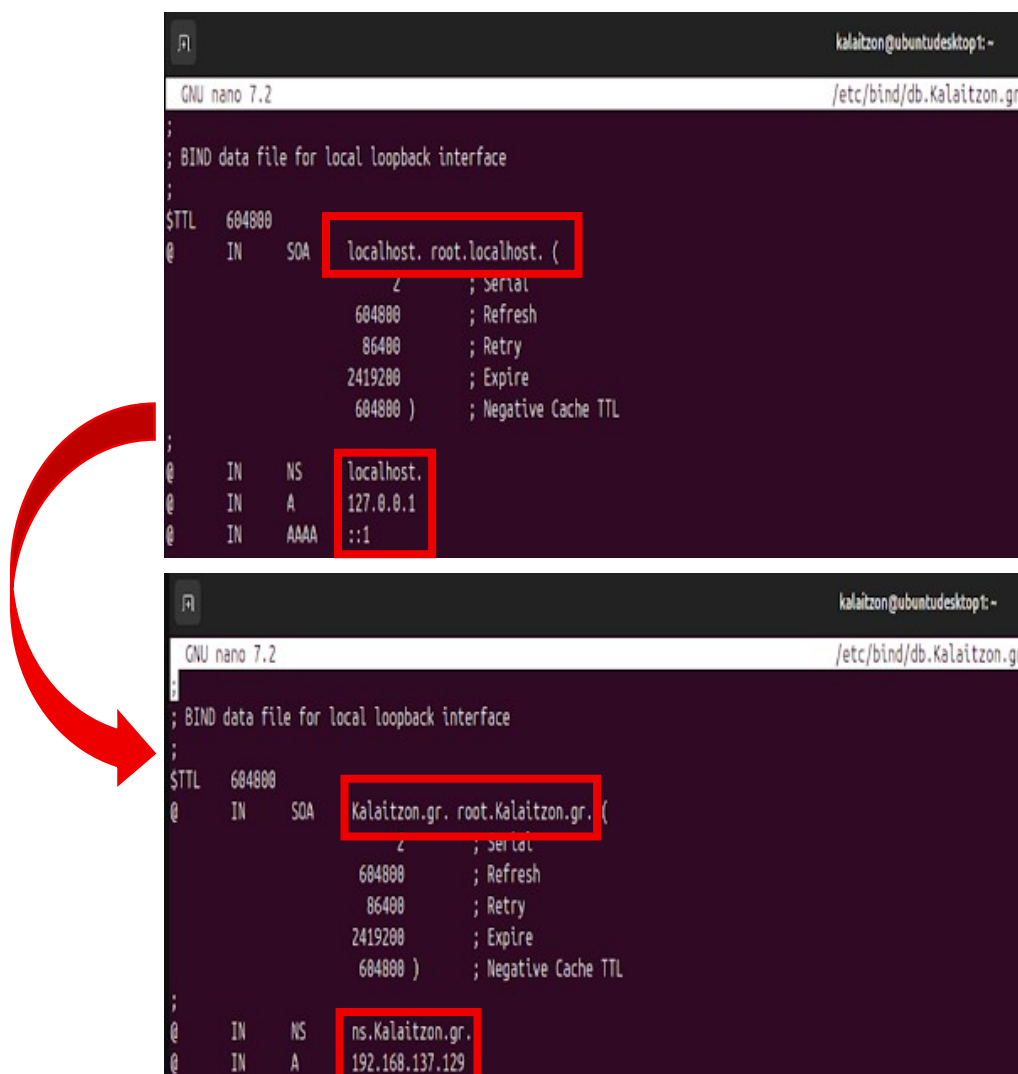
because when the user, e.g. from Windows, asks the server, for example, what YouTube's IP is, it will forward the question to Google's IP and in this way the server will provide the answer. Under other circumstances our server would be isolated and would not know anything other than our own domain.

Then we will find the host's IP with the command `ip a`. The Ubuntu IP is 192.168.137.129

Next, we proceed to create our own Local Zone. With the command `sudo nano /etc/bind/named.conf.local` we edit the settings file to declare the 'Kalaitzon.gr' domain. Inside the file, we set the zone type as master, thus declaring that our server will be the Authoritative Name Server for this domain. Creating the Local Zone is necessary

because the 'Kalaitzon.gr' domain is private and does not exist on the internet. Consequently, DNS servers, such as Google's, do not have records for it and are unable to resolve it. Through the local zone, we set our own server as the exclusive Authoritative Source for this domain. Thus, we achieve name resolution locally, without depending on external providers, essentially creating an internal network. At the same time, with the file directive, we link the zone to the db.Kalaitzon.gr file (which we will create below). This will be the data file that includes all the necessary information for our domain. In other words, we separate the resolution process. If a client requests an address outside our domain, the Forwarding we configured earlier is activated. Conversely, for queries concerning 'Kalaitzon.gr', the server does not turn to Google's IP, but draws the information directly from this local file.

Now we will create the file we declared in the zone above. With the command `sudo cp /etc/bind/db.local /etc/bind/db.Kalaitzon.gr` we will copy the template file that exists from localhost to our own (db.Kalaitzon.gr) and then edit it with the command `sudo nano /etc/bind/db.Kalaitzon.gr`. Wherever it says localhost, we replace it with Kalaitzon.gr (our domain), and wherever we see an IP we put that of Ubuntu (192.168.137.129).



```
kalaitzon@ubuntudesktop: ~  
GNU nano 7.2 /etc/bind/db.Kalaitzon.gr  
; BIND data file for local loopback interface  
;  
$TTL 604800  
@ IN SOA localhost. root.localhost. (  
                2      ; Serial  
                604800 ; Refresh  
                86400  ; Retry  
                2419200 ; Expire  
                604800 ) ; Negative Cache TTL  
;  
@ IN NS localhost.  
@ IN A 127.0.0.1  
@ IN AAAA ::1  
  
kalaitzon@ubuntudesktop: ~  
GNU nano 7.2 /etc/bind/db.Kalaitzon.gr  
; BIND data file for local loopback interface  
;  
$TTL 604800  
@ IN SOA Kalaitzon.gr. root.Kalaitzon.gr. (  
                2      ; Serial  
                604800 ; Refresh  
                86400  ; Retry  
                2419200 ; Expire  
                604800 ) ; Negative Cache TTL  
;  
@ IN NS ns.Kalaitzon.gr.  
@ IN A 192.168.137.129
```

```

kalaitzon@ubuntudesktop1:~$ sudo systemctl restart bind9
kalaitzon@ubuntudesktop1:~$ sudo systemctl status bind9
● named.service - BIND Domain Name Server
   Loaded: loaded (/usr/lib/systemd/system/named.service; enabled; preset: enabled)
   Active: active (running) since Sat 2025-11-22 01:38:52 EET; 3s ago
     Docs: man:named(8)
    Main PID: 13461 (named)
    Status: "running"
      Tasks: 10 (limit: 9374)
    Memory: 23.9M (peak: 24.4M)
       CPU: 297ms
    CGroup: /system.slice/named.service
           └─13461 /usr/sbin/named -f -u bind

Nov 22 01:38:54 ubuntudesktop1.kalaitzon named[13461]: network unreachable resolving './DNSKEY/IN': 2001:dc3::35#53
Nov 22 01:38:54 ubuntudesktop1.kalaitzon named[13461]: network unreachable resolving './DNSKEY/IN': 2001:503:ba3e::2:30#53
Nov 22 01:38:54 ubuntudesktop1.kalaitzon named[13461]: network unreachable resolving './DNSKEY/IN': 2001:503:c27::2:30#53
Nov 22 01:38:54 ubuntudesktop1.kalaitzon named[13461]: network unreachable resolving './DNSKEY/IN': 2001:500:2::c#53
Nov 22 01:38:54 ubuntudesktop1.kalaitzon named[13461]: network unreachable resolving './DNSKEY/IN': 2001:7fd::1#53
Nov 22 01:38:54 ubuntudesktop1.kalaitzon named[13461]: network unreachable resolving './DNSKEY/IN': 2001:500:1::53#53
Nov 22 01:38:54 ubuntudesktop1.kalaitzon named[13461]: network unreachable resolving './DNSKEY/IN': 2801:1b8:10::b#53
Nov 22 01:38:54 ubuntudesktop1.kalaitzon named[13461]: network unreachable resolving './DNSKEY/IN': 2001:7fe::53#53
Nov 22 01:38:54 ubuntudesktop1.kalaitzon named[13461]: managed-keys-zone: Key 20326 for zone . is now trusted (acceptance timer complete)
Nov 22 01:38:54 ubuntudesktop1.kalaitzon named[13461]: managed-keys-zone: Key 38696 for zone . is now trusted (acceptance timer complete)
kalaitzon@ubuntudesktop1:~$

```

We make the standard keystrokes to save the file and proceed to restart the server with the command `sudo systemctl restart bind9`. If it produces no error message, then it means it worked. To check that everything went well and it is working, we will use the command `sudo systemctl status bind9` and observe the line that says Active: active (running)

To confirm the correct operation of the DNS Server, a second virtual machine with a Windows 10 operating system was used, which acted as the Client. Initially, an attempt was made to resolve the domain name `www.kalaitzon.gr` via the `nslookup` command. This attempt failed (displaying a "Non-existent domain" message), because the client was configured to use the network's default DNS server (VMware's Gateway, 192.168.137.2), which did not have records for our local zone. To address this, the client's Network Adapter Settings were modified. Specifically, in the TCP/IPv4 protocol, the IP address of our Ubuntu Server (192.168.137.129) was statically set as the Preferred DNS Server. After applying the settings, the command `nslookup www.kalaitzon.gr` was executed again. This time, the client turned directly to our own DNS Server, which responded successfully, resolving the domain name to the correct IP address (192.168.137.129), thus confirming that the Authoritative Server has been configured correctly.

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